

Distributivity If a , b , and c belong to $\mathbf{Z}_m$, then $a \cdot_m (b +_m c) = (a \cdot_m b) +_m (a \cdot_m c)$ and $(a +_m b) \cdot_m c = (a \cdot_m c) +_m (b \cdot_m c)$.

These properties follow from the properties we have developed for congruences and remainders modulo m , together with the properties of integers; we leave their proofs as Exercises 48–50. Note that we have listed the property that every element of $\mathbf{Z}_m$ has an additive inverse, but no analogous property for multiplicative inverses has been included. This is because multiplicative inverses do not always exist modulo m . For instance, there is no multiplicative inverse of 2 modulo 6, as the reader can verify. We will return to the question of when an integer has a multiplicative inverse modulo m later in this chapter.

Remark: Because $\mathbf{Z}_m$ with the operations of addition and multiplication modulo m satisfies the properties listed, $\mathbf{Z}_m$ with modular addition is said to be a **commutative group** and $\mathbf{Z}_m$ with both of these operations is said to be a **commutative ring**. Note that the set of integers with ordinary addition and multiplication also forms a commutative ring. Groups and rings are studied in courses that cover abstract algebra.

Remark: In Exercise 36, and in later sections, we will use the notations $+$ and $\cdot$ for $+_m$ and $\cdot_m$ without the subscript m on the symbol for the operator whenever we work with $\mathbf{Z}_m$.

Exercises

- Does 17 divide each of these numbers?
a) 68 b) 84 c) 357 d) 1001
- Prove that if a is an integer other than 0, then
a) 1 divides a . b) a divides 0.
- Prove that part (ii) of Theorem 1 is true.
- Prove that part (iii) of Theorem 1 is true.
- Show that if $a \mid b$ and $b \mid a$, where a and b are integers, then $a = b$ or $a = -b$.
- Show that if a, b, c , and d are integers, where $a \neq 0$, such that $a \mid c$ and $b \mid d$, then $ab \mid cd$.
- Show that if a, b , and c are integers, where $a \neq 0$ and $c \neq 0$, such that $ac \mid bc$, then $a \mid b$.
- Prove or disprove that if $a \mid bc$, where a, b , and c are positive integers and $a \neq 0$, then $a \mid b$ or $a \mid c$.
- Prove that if a and b are integers and a divides b , then a is odd or b is even.
- Prove that if a and b are nonzero integers, a divides b , and $a + b$ is odd, then a is odd.
- Prove that if a is an integer that is not divisible by 3, then $(a + 1)(a + 2)$ is divisible by 3.
- Prove that if a is a positive integer, then 4 does not divide $a^2 + 2$.
- What are the quotient and remainder when
a) 19 is divided by 7? b) -111 is divided by 11?
c) 789 is divided by 23? d) 1001 is divided by 13?
e) 0 is divided by 19? f) 3 is divided by 5?
g) -1 is divided by 3? h) 4 is divided by 1?
- What are the quotient and remainder when
a) 44 is divided by 8?
b) 777 is divided by 21?
c) -123 is divided by 19?
- -1 is divided by 23?
e) -2002 is divided by 87?
f) 0 is divided by 17?
g) 1,234,567 is divided by 1001?
h) -100 is divided by 101?
- What time does a 12-hour clock read
a) 80 hours after it reads 11:00?
b) 40 hours before it reads 12:00?
c) 100 hours after it reads 6:00?
- What time does a 24-hour clock read
a) 100 hours after it reads 2:00?
b) 45 hours before it reads 12:00?
c) 168 hours after it reads 19:00?
- Suppose that a and b are integers, $a \equiv 4 \pmod{13}$, and $b \equiv 9 \pmod{13}$. Find the integer c with $0 \leq c \leq 12$ such that
a) $c \equiv 9a \pmod{13}$.
b) $c \equiv 11b \pmod{13}$.
c) $c \equiv a + b \pmod{13}$.
d) $c \equiv 2a + 3b \pmod{13}$.
e) $c \equiv a^2 + b^2 \pmod{13}$.
f) $c \equiv a^3 - b^3 \pmod{13}$.
- Suppose that a and b are integers, $a \equiv 11 \pmod{19}$, and $b \equiv 3 \pmod{19}$. Find the integer c with $0 \leq c \leq 18$ such that
a) $c \equiv 13a \pmod{19}$.
b) $c \equiv 8b \pmod{19}$.
c) $c \equiv a - b \pmod{19}$.
d) $c \equiv 7a + 3b \pmod{19}$.
e) $c \equiv 2a^2 + 3b^2 \pmod{19}$.
f) $c \equiv a^3 + 4b^3 \pmod{19}$.

19. Show that if a and d are positive integers, then $(-a) \operatorname{div} d = -a \operatorname{div} d$ if and only if d divides a .
20. Prove or disprove that if a , b , and d are integers with $d > 0$, then $(a + b) \operatorname{div} d = a \operatorname{div} d + b \operatorname{div} d$.
21. Let m be a positive integer. Show that $a \equiv b \pmod{m}$ if $a \bmod m = b \bmod m$.
22. Let m be a positive integer. Show that $a \bmod m = b \bmod m$ if $a \equiv b \pmod{m}$.
23. Show that if n and k are positive integers, then $\lceil n/k \rceil = \lfloor (n-1)/k \rfloor + 1$.
24. Show that if a is an integer and d is an integer greater than 1, then the quotient and remainder obtained when a is divided by d are $\lfloor a/d \rfloor$ and $a - d\lfloor a/d \rfloor$, respectively.
25. Find a formula for the integer with smallest absolute value that is congruent to an integer a modulo m , where m is a positive integer.
26. Evaluate these quantities.

a) $-17 \bmod 2$	b) $144 \bmod 7$
c) $-101 \bmod 13$	d) $199 \bmod 19$
27. Evaluate these quantities.

a) $13 \bmod 3$	b) $-97 \bmod 11$
c) $155 \bmod 19$	d) $-221 \bmod 23$
28. Find $a \operatorname{div} m$ and $a \bmod m$ when

a) $a = -111, m = 99$.
b) $a = -9999, m = 101$.
c) $a = 10299, m = 999$.
d) $a = 123456, m = 1001$.
29. Find $a \operatorname{div} m$ and $a \bmod m$ when

a) $a = 228, m = 119$.
b) $a = 9009, m = 223$.
c) $a = -10101, m = 333$.
d) $a = -765432, m = 38271$.
30. Find the integer a such that

a) $a \equiv 43 \pmod{23}$ and $-22 \leq a \leq 0$.
b) $a \equiv 17 \pmod{29}$ and $-14 \leq a \leq 14$.
c) $a \equiv -11 \pmod{21}$ and $90 \leq a \leq 110$.
31. Find the integer a such that

a) $a \equiv -15 \pmod{27}$ and $-26 \leq a \leq 0$.
b) $a \equiv 24 \pmod{31}$ and $-15 \leq a \leq 15$.
c) $a \equiv 99 \pmod{41}$ and $100 \leq a \leq 140$.
32. List five integers that are congruent to 4 modulo 12.
33. List all integers between -100 and 100 that are congruent to -1 modulo 25.
34. Decide whether each of these integers is congruent to 3 modulo 7.

a) 37	b) 66
c) -17	d) -67
35. Decide whether each of these integers is congruent to 5 modulo 17.

a) 80	b) 103
c) -29	d) -122
36. Find each of these values.

a) $(177 \bmod 31 + 270 \bmod 31) \bmod 31$	b) $(177 \bmod 31 \cdot 270 \bmod 31) \bmod 31$
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37. Find each of these values.

a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$	b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$
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38. Find each of these values.

a) $(19^2 \bmod 41) \bmod 9$
b) $(32^3 \bmod 13)^2 \bmod 11$
c) $(7^3 \bmod 23)^2 \bmod 31$
d) $(21^2 \bmod 15)^3 \bmod 22$
39. Find each of these values.

a) $(99^2 \bmod 32)^3 \bmod 15$
b) $(3^4 \bmod 17)^2 \bmod 11$
c) $(19^3 \bmod 23)^2 \bmod 31$
d) $(89^3 \bmod 79)^4 \bmod 26$
40. Show that if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, where a, b, c, d , and m are integers with $m \geq 2$, then $a - c \equiv b - d \pmod{m}$.
41. Show that if $n \mid m$, where n and m are integers greater than 1, and if $a \equiv b \pmod{m}$, where a and b are integers, then $a \equiv b \pmod{n}$.
42. Show that if a, b, c , and m are integers such that $m \geq 2$, $c > 0$, and $a \equiv b \pmod{m}$, then $ac \equiv bc \pmod{mc}$.
43. Find counterexamples to each of these statements about congruences.

a) If $ac \equiv bc \pmod{m}$, where a, b, c , and m are integers with $m \geq 2$, then $a \equiv b \pmod{m}$.
b) If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, where a, b, c, d , and m are integers with c and d positive and $m \geq 2$, then $a^c \equiv b^d \pmod{m}$.
44. Show that if n is an integer then $n^2 \equiv 0$ or $1 \pmod{4}$.
45. Use Exercise 44 to show that if m is a positive integer of the form $4k + 3$ for some nonnegative integer k , then m is not the sum of the squares of two integers.
46. Prove that if n is an odd positive integer, then $n^2 \equiv 1 \pmod{8}$.
47. Show that if a, b, k , and m are integers such that $k \geq 1$, $m \geq 2$, and $a \equiv b \pmod{m}$, then $a^k \equiv b^k \pmod{m}$.
48. Show that $\mathbf{Z}_m$ with addition modulo m , where $m \geq 2$ is an integer, satisfies the closure, associative, and commutative properties, 0 is an additive identity, and for every nonzero $a \in \mathbf{Z}_m$, $m - a$ is an inverse of a modulo m .
49. Show that $\mathbf{Z}_m$ with multiplication modulo m , where $m \geq 2$ is an integer, satisfies the closure, associative, and commutativity properties, and 1 is a multiplicative identity.
50. Show that the distributive property of multiplication over addition holds for $\mathbf{Z}_m$, where $m \geq 2$ is an integer.
51. Write out the addition and multiplication tables for $\mathbf{Z}_5$ (where by addition and multiplication we mean $+_5$ and $\cdot_5$).
52. Write out the addition and multiplication tables for $\mathbf{Z}_6$ (where by addition and multiplication we mean $+_6$ and $\cdot_6$).
53. Determine whether each of the functions $f(a) = a \operatorname{div} d$ and $g(a) = a \bmod d$, where d is a fixed positive integer, from the set of integers to the set of integers, is one-to-one, and determine whether each of these functions is onto.

Exercises

- Convert the decimal expansion of each of these integers to a binary expansion.
a) 231 b) 4532 c) 97644
- Convert the decimal expansion of each of these integers to a binary expansion.
a) 321 b) 1023 c) 100632
- Convert the binary expansion of each of these integers to a decimal expansion.
a) $(1\ 1111)_2$ b) $(10\ 0000\ 0001)_2$
c) $(1\ 0101\ 0101)_2$ d) $(110\ 1001\ 0001\ 0000)_2$
- Convert the binary expansion of each of these integers to a decimal expansion.
a) $(1\ 1011)_2$ b) $(10\ 1011\ 0101)_2$
c) $(11\ 1011\ 1110)_2$ d) $(111\ 1100\ 0001\ 1111)_2$
- Convert the octal expansion of each of these integers to a binary expansion.
a) $(572)_8$ b) $(1604)_8$
c) $(423)_8$ d) $(2417)_8$
- Convert the binary expansion of each of these integers to an octal expansion.
a) $(1111\ 0111)_2$
b) $(1010\ 1010\ 1010)_2$
c) $(111\ 0111\ 0111\ 0111)_2$
d) $(101\ 0101\ 0101\ 0101)_2$
- Convert the hexadecimal expansion of each of these integers to a binary expansion.
a) $(80E)_{16}$ b) $(135AB)_{16}$
c) $(ABBA)_{16}$ d) $(DEFACED)_{16}$
- Convert $(BADFACED)_{16}$ from its hexadecimal expansion to its binary expansion.
- Convert $(ABCDEF)_{16}$ from its hexadecimal expansion to its binary expansion.
- Convert each of the integers in Exercise 6 from a binary expansion to a hexadecimal expansion.
- Convert $(1011\ 0111\ 1011)_2$ from its binary expansion to its hexadecimal expansion.
- Convert $(1\ 1000\ 0110\ 0011)_2$ from its binary expansion to its hexadecimal expansion.
- Show that the hexadecimal expansion of a positive integer can be obtained from its binary expansion by grouping together blocks of four binary digits, adding initial zeros if necessary, and translating each block of four binary digits into a single hexadecimal digit.
- Show that the binary expansion of a positive integer can be obtained from its hexadecimal expansion by translating each hexadecimal digit into a block of four binary digits.
- Show that the octal expansion of a positive integer can be obtained from its binary expansion by grouping together blocks of three binary digits, adding initial zeros if necessary, and translating each block of three binary digits into a single octal digit.
- Show that the binary expansion of a positive integer can be obtained from its octal expansion by translating each octal digit into a block of three binary digits.
- Convert $(7345321)_8$ to its binary expansion and $(10\ 1011\ 1011)_2$ to its octal expansion.
- Give a procedure for converting from the hexadecimal expansion of an integer to its octal expansion using binary notation as an intermediate step.
- Give a procedure for converting from the octal expansion of an integer to its hexadecimal expansion using binary notation as an intermediate step.
- Explain how to convert from binary to base 64 expansions and from base 64 expansions to binary expansions and from octal to base 64 expansions and from base 64 expansions to octal expansions.
- Find the sum and the product of each of these pairs of numbers. Express your answers as a binary expansion.
a) $(100\ 0111)_2, (111\ 0111)_2$
b) $(1110\ 1111)_2, (1011\ 1101)_2$
c) $(10\ 1010\ 1010)_2, (1\ 1111\ 0000)_2$
d) $(10\ 0000\ 0001)_2, (11\ 1111\ 1111)_2$
- Find the sum and product of each of these pairs of numbers. Express your answers as a base 3 expansion.
a) $(112)_3, (210)_3$
b) $(2112)_3, (12021)_3$
c) $(20001)_3, (1111)_3$
d) $(120021)_3, (2002)_3$
- Find the sum and product of each of these pairs of numbers. Express your answers as an octal expansion.
a) $(763)_8, (147)_8$
b) $(6001)_8, (272)_8$
c) $(1111)_8, (777)_8$
d) $(54321)_8, (3456)_8$
- Find the sum and product of each of these pairs of numbers. Express your answers as a hexadecimal expansion.
a) $(1AE)_{16}, (BBC)_{16}$
b) $(20CBA)_{16}, (A01)_{16}$
c) $(ABCDE)_{16}, (1111)_{16}$
d) $(E0000E)_{16}, (BAAA)_{16}$
- Use Algorithm 5 to find $7^{644} \bmod 645$.
- Use Algorithm 5 to find $11^{644} \bmod 645$.
- Use Algorithm 5 to find $3^{2003} \bmod 99$.
- Use Algorithm 5 to find $123^{1001} \bmod 101$.
- Show that every positive integer can be represented uniquely as the sum of distinct powers of 2. [Hint: Consider binary expansions of integers.]
- It can be shown that every integer can be uniquely represented in the form
$$e_k 3^k + e_{k-1} 3^{k-1} + \cdots + e_1 3 + e_0,$$
where $e_j = -1, 0, \text{ or } 1$ for $j = 0, 1, 2, \dots, k$. Expansions of this type are called **balanced ternary expansions**. Find the balanced ternary expansions of
a) 5. b) 13. c) 37. d) 79.

31. Show that a positive integer is divisible by 3 if and only if the sum of its decimal digits is divisible by 3.
32. Show that a positive integer is divisible by 11 if and only if the difference of the sum of its decimal digits in even-numbered positions and the sum of its decimal digits in odd-numbered positions is divisible by 11.
33. Show that a positive integer is divisible by 3 if and only if the difference of the sum of its binary digits in even-numbered positions and the sum of its binary digits in odd-numbered positions is divisible by 3.
34. Determine how we can use the decimal expansion of an integer n to determine whether n is divisible by
a) 2 b) 5 c) 10
35. Determine how we can use the decimal expansion of an integer n to determine whether n is divisible by
a) 4 b) 25 c) 20
36. Suppose that n and b are positive integers with $b \geq 2$ and the base b expansion of n is $n = (a_m a_{m-1} \dots a_1 a_0)_b$. Find the base b expansion of
a) bn . b) $b^2 n$;
c) $\lfloor n/b \rfloor$, d) $\lfloor n/b^2 \rfloor$.
37. Prove that if n and b are positive integers with $b \geq 2$ the base b representation of n has $\lfloor \log_b n \rfloor + 1$ digits.
38. Find the decimal expansion of the number with the n -digit base seven expansion $(111 \dots 111)_7$ (with n 1's). [Hint: Use the formula for the sum of the terms of a geometric progression.]
39. Find the decimal expansion of the number with the $3n$ bit binary expansion $(101101 \dots 101101)_2$ (so that the binary expansion is made of n copies of 101). [Hint: Use the formula for the sum of the terms of a geometric progression.]
- One's complement** representations of integers are used to simplify computer arithmetic. To represent positive and negative integers with absolute value less than 2^{n-1} , a total of n bits is used. The leftmost bit is used to represent the sign. A 0 bit in this position is used for positive integers, and a 1 bit in this position is used for negative integers. For positive integers, the remaining bits are identical to the binary expansion of the integer. For negative integers, the remaining bits are obtained by first finding the binary expansion of the absolute value of the integer, and then taking the complement of each of these bits, where the complement of a 1 is a 0 and the complement of a 0 is a 1.
40. Find the one's complement representations, using bit strings of length six, of the following integers.
a) 22 b) 31 c) -7 d) -19
41. What integer does each of the following one's complement representations of length five represent?
a) 11001 b) 01101
c) 10001 d) 11111
42. If m is a positive integer less than 2^{n-1} , how is the one's complement representation of $-m$ obtained from the one's complement of m , when bit strings of length n are used?

43. How is the one's complement representation of the sum of two integers obtained from the one's complement representations of these integers?
44. How is the one's complement representation of the difference of two integers obtained from the one's complement representations of these integers?
45. Show that the integer m with one's complement representation $(a_{n-1} a_{n-2} \dots a_1 a_0)$ can be found using the equation $m = -a_{n-1}(2^{n-1} - 1) + a_{n-2}2^{n-2} + \dots + a_1 \cdot 2 + a_0$.

Two's complement representations of integers are also used to simplify computer arithmetic and are used more commonly than one's complement representations. To represent an integer x with $-2^{n-1} \leq x \leq 2^{n-1} - 1$ for a specified positive integer n , a total of n bits is used. The leftmost bit is used to represent the sign. A 0 bit in this position is used for positive integers, and a 1 bit in this position is used for negative integers, just as in one's complement expansions. For a positive integer, the remaining bits are identical to the binary expansion of the integer. For a negative integer, the remaining bits are the bits of the binary expansion of $2^{n-1} - |x|$. Two's complement expansions of integers are often used by computers because addition and subtraction of integers can be performed easily using these expansions, where these integers can be either positive or negative.

46. Answer Exercise 40, but this time find the two's complement expansion using bit strings of length six.
47. Answer Exercise 41 if each expansion is a two's complement expansion of length five.
48. Answer Exercise 42 for two's complement expansions.
49. Answer Exercise 43 for two's complement expansions.
50. Answer Exercise 44 for two's complement expansions.
51. Show that the integer m with two's complement representation $(a_{n-1} a_{n-2} \dots a_1 a_0)$ can be found using the equation $m = -a_{n-1} \cdot 2^{n-1} + a_{n-2}2^{n-2} + \dots + a_1 \cdot 2 + a_0$.
52. Give a simple algorithm for forming the two's complement representation of an integer from its one's complement representation.
53. Sometimes integers are encoded by using four-digit binary expansions to represent each decimal digit. This produces the **binary coded decimal** form of the integer. For instance, 791 is encoded in this way by 011110010001. How many bits are required to represent a number with n decimal digits using this type of encoding?

A **Cantor expansion** is a sum of the form

$$a_n n! + a_{n-1}(n-1)! + \dots + a_2 2! + a_1 1!,$$

where a_i is an integer with $0 \leq a_i \leq i$ for $i = 1, 2, \dots, n$.

54. Find the Cantor expansions of
a) 2. b) 7.
c) 19. d) 87.
e) 1000. f) 1,000,000.

- *55. Describe an algorithm that finds the Cantor expansion of an integer.
- *56. Describe an algorithm to add two integers from their Cantor expansions.
- 57. Add $(10111)_2$ and $(11010)_2$ by working through each step of the algorithm for addition given in the text.
- 58. Multiply $(1110)_2$ and $(1010)_2$ by working through each step of the algorithm for multiplication given in the text.
- 59. Describe an algorithm for finding the difference of two binary expansions.
- 60. Estimate the number of bit operations used to subtract two binary expansions.
- 61. Devise an algorithm that, given the binary expansions of the integers a and b , determines whether $a > b$, $a = b$, or $a < b$.
- 62. How many bit operations does the comparison algorithm from Exercise 61 use when the larger of a and b has n bits in its binary expansion?
- 63. Estimate the complexity of Algorithm 1 for finding the base b expansion of an integer n in terms of the number of divisions used.
- *64. Show that Algorithm 5 uses $O((\log m)^2 \log n)$ bit operations to find $b^n \bmod m$.
- 65. Show that Algorithm 4 uses $O(q \log a)$ bit operations, assuming that $a > d$.

4.3 Primes and Greatest Common Divisors

4.3.1 Introduction

In Section 4.1 we studied the concept of divisibility of integers. One important concept based on divisibility is that of a prime number. A prime is an integer greater than 1 that is divisible by no positive integers other than 1 and itself. The study of prime numbers goes back to ancient times. Thousands of years ago it was known that there are infinitely many primes; the proof of this fact, found in the works of Euclid, is famous for its elegance and beauty.

We will discuss the distribution of primes among the integers. We will describe some of the results about primes found by mathematicians in the last 400 years. In particular, we will introduce an important theorem, the fundamental theorem of arithmetic. This theorem, which asserts that every positive integer can be written uniquely as the product of primes in nondecreasing order, has many interesting consequences. We will also discuss some of the many old conjectures about primes that remain unsettled today.

Primes have become essential in modern cryptographic systems, and we will develop some of their properties important in cryptography. For example, finding large primes is essential in modern cryptography. The length of time required to factor large integers into their prime factors is the basis for the strength of some important modern cryptographic systems.

In this section we will also study the greatest common divisor of two integers, as well as the least common multiple of two integers. We will develop an important algorithm for computing greatest common divisors, called the Euclidean algorithm.

4.3.2 Primes

Every integer greater than 1 is divisible by at least two integers, because a positive integer is divisible by 1 and by itself. Positive integers that have exactly two different positive integer factors are called **primes**.

Definition 1

An integer p greater than 1 is called *prime* if the only positive factors of p are 1 and p . A positive integer that is greater than 1 and is not prime is called *composite*.

Remark: The integer 1 is not prime, because it has only one positive factor. Note also that an integer n is composite if and only if there exists an integer a such that $a \mid n$ and $1 < a < n$.

We can now show that a factorization of an integer into primes is unique. That is, we will show that every integer can be written as the product of primes in nondecreasing order in at most one way. This is part of the fundamental theorem of arithmetic. We will prove the other part, that every integer has a factorization into primes, in Section 5.2.

Proof (of the uniqueness of the prime factorization of a positive integer): We will use a proof by contradiction. Suppose that the positive integer n can be written as the product of primes in two different ways, say, $n = p_1 p_2 \cdots p_s$ and $n = q_1 q_2 \cdots q_t$, where each p_i and q_j is prime such that $p_1 \leq p_2 \leq \cdots \leq p_s$ and $q_1 \leq q_2 \leq \cdots \leq q_t$.

When we remove all common primes from the two factorizations, we have

$$p_{i_1} p_{i_2} \cdots p_{i_u} = q_{j_1} q_{j_2} \cdots q_{j_v},$$

where no prime occurs on both sides of this equation and u and v are positive integers. By Lemma 3 it follows that p_{i_1} divides q_{j_k} for some k . Because no prime divides another prime, this is impossible. Consequently, there can be at most one factorization of n into primes in nondecreasing order. $\triangleleft$

Lemma 2 can also be used to prove a result about dividing both sides of a congruence by the same integer. We have shown (Theorem 5 in Section 4.1) that we can multiply both sides of a congruence by the same integer. However, dividing both sides of a congruence by an integer does not always produce a valid congruence, as Example 19 shows.

EXAMPLE 19 The congruence $14 \equiv 8 \pmod{6}$ holds, but both sides of this congruence cannot be divided by 2 to produce a valid congruence because $14/2 = 7$ and $8/2 = 4$, but $7 \not\equiv 4 \pmod{6}$. $\triangleleft$

Although we cannot divide both sides of a congruence by any integer to produce a valid congruence, we can if this integer is relatively prime to the modulus. Theorem 7 establishes this important fact. We use Lemma 2 in the proof.

THEOREM 7 Let m be a positive integer and let a , b , and c be integers. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Proof: Because $ac \equiv bc \pmod{m}$, $m \mid ac - bc = c(a - b)$. By Lemma 2, because $\gcd(c, m) = 1$, it follows that $m \mid a - b$. We conclude that $a \equiv b \pmod{m}$. $\triangleleft$

Exercises

- Determine whether each of these integers is prime.

a) 21	b) 29
c) 71	d) 97
e) 111	f) 143
- Determine whether each of these integers is prime.

a) 19	b) 27
c) 93	d) 101
e) 107	f) 113
- Find the prime factorization of each of these integers.

a) 88	b) 126	c) 729
d) 1001	e) 1111	f) 909,090
- Find the prime factorization of each of these integers.

a) 39	b) 81	c) 101
d) 143	e) 289	f) 899
- Find the prime factorization of $10!$.
- *6. How many zeros are there at the end of $100!$?
- Express in pseudocode the trial division algorithm for determining whether an integer is prime.
- Express in pseudocode the algorithm described in the text for finding the prime factorization of an integer.
- Show that $a^m + 1$ is composite if a and m are integers greater than 1 and m is odd. [Hint: Show that $x + 1$ is a factor of the polynomial $x^m + 1$ if m is odd.]

10. Show that if $2^m + 1$ is an odd prime, then $m = 2^n$ for some nonnegative integer n . [Hint: First show that the polynomial identity $x^m + 1 = (x^k + 1)(x^{k(t-1)} - x^{k(t-2)} + \cdots - x^k + 1)$ holds, where $m = kt$ and t is odd.]
- *11. Show that $\log_2 3$ is an irrational number. Recall that an irrational number is a real number x that cannot be written as the ratio of two integers.
12. Prove that for every positive integer n , there are n consecutive composite integers. [Hint: Consider the n consecutive integers starting with $(n+1)! + 2$.]
- *13. Prove or disprove that there are three consecutive odd positive integers that are primes, that is, odd primes of the form p , $p+2$, and $p+4$.
14. Which positive integers less than 12 are relatively prime to 12?
15. Which positive integers less than 30 are relatively prime to 30?
16. Determine whether the integers in each of these sets are pairwise relatively prime.
 a) 21, 34, 55 b) 14, 17, 85
 c) 25, 41, 49, 64 d) 17, 18, 19, 23
17. Determine whether the integers in each of these sets are pairwise relatively prime.
 a) 11, 15, 19 b) 14, 15, 21
 c) 12, 17, 31, 37 d) 7, 8, 9, 11
18. We call a positive integer **perfect** if it equals the sum of its positive divisors other than itself.
 a) Show that 6 and 28 are perfect.
 b) Show that $2^{p-1}(2^p - 1)$ is a perfect number when $2^p - 1$ is prime.
19. Show that if $2^n - 1$ is prime, then n is prime. [Hint: Use the identity $2^{ab} - 1 = (2^a - 1) \cdot (2^{a(b-1)} + 2^{a(b-2)} + \cdots + 2^a + 1)$.]
20. Determine whether each of these integers is prime, verifying some of Mersenne's claims.
 a) $2^7 - 1$ b) $2^9 - 1$
 c) $2^{11} - 1$ d) $2^{13} - 1$
- The value of the **Euler ϕ -function** at the positive integer n is defined to be the number of positive integers less than or equal to n that are relatively prime to n . For instance, $\phi(6) = 2$ because of the positive integers less or equal to 6, only 1 and 5 are relatively prime to 6. [Note: ϕ is the Greek letter phi.]
21. Find these values of the Euler ϕ -function.
 a) $\phi(4)$ b) $\phi(10)$ c) $\phi(13)$
22. Show that n is prime if and only if $\phi(n) = n - 1$.
23. What is the value of $\phi(p^k)$ when p is prime and k is a positive integer?
24. What are the greatest common divisors of these pairs of integers?
 a) $2^2 \cdot 3^3 \cdot 5^5, 2^5 \cdot 3^3 \cdot 5^2$
 b) $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13, 2^{11} \cdot 3^9 \cdot 11 \cdot 17^{14}$
 c) $17, 17^{17}$ d) $2^2 \cdot 7, 5^3 \cdot 13$
 e) $0, 5$ f) $2 \cdot 3 \cdot 5 \cdot 7, 2 \cdot 3 \cdot 5 \cdot 7$
25. What are the greatest common divisors of these pairs of integers?
 a) $3^7 \cdot 5^3 \cdot 7^3, 2^{11} \cdot 3^5 \cdot 5^9$
 b) $11 \cdot 13 \cdot 17, 2^9 \cdot 3^7 \cdot 5^5 \cdot 7^3$
 c) $23^{31}, 23^{17}$
 d) $41 \cdot 43 \cdot 53, 41 \cdot 43 \cdot 53$
 e) $3^{13} \cdot 5^{17}, 2^{12} \cdot 7^{21}$
 f) $1111, 0$
26. What is the least common multiple of each pair in Exercise 24?
27. What is the least common multiple of each pair in Exercise 25?
28. Find $\gcd(1000, 625)$ and $\text{lcm}(1000, 625)$ and verify that $\gcd(1000, 625) \cdot \text{lcm}(1000, 625) = 1000 \cdot 625$.
29. Find $\gcd(92928, 123552)$ and $\text{lcm}(92928, 123552)$, and verify that $\gcd(92928, 123552) \cdot \text{lcm}(92928, 123552) = 92928 \cdot 123552$. [Hint: First find the prime factorizations of 92928 and 123552.]
30. If the product of two integers is $2^7 3^8 5^2 7^{11}$ and their greatest common divisor is $2^3 3^4 5$, what is their least common multiple?
31. Show that if a and b are positive integers, then $ab = \gcd(a, b) \cdot \text{lcm}(a, b)$. [Hint: Use the prime factorizations of a and b and the formulae for $\gcd(a, b)$ and $\text{lcm}(a, b)$ in terms of these factorizations.]
32. Use the Euclidean algorithm to find
 a) $\gcd(1, 5)$. b) $\gcd(100, 101)$.
 c) $\gcd(123, 277)$. d) $\gcd(1529, 14039)$.
 e) $\gcd(1529, 14038)$. f) $\gcd(11111, 111111)$.
33. Use the Euclidean algorithm to find
 a) $\gcd(12, 18)$. b) $\gcd(111, 201)$.
 c) $\gcd(1001, 1331)$. d) $\gcd(12345, 54321)$.
 e) $\gcd(1000, 5040)$. f) $\gcd(9888, 6060)$.
34. How many divisions are required to find $\gcd(21, 34)$ using the Euclidean algorithm?
35. How many divisions are required to find $\gcd(34, 55)$ using the Euclidean algorithm?
- *36. Show that if a and b are both positive integers, then $(2^a - 1) \bmod (2^b - 1) = 2^{a \bmod b} - 1$.
-  *37. Use Exercise 36 to show that if a and b are positive integers, then $\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a, b)} - 1$. [Hint: Show that the remainders obtained when the Euclidean algorithm is used to compute $\gcd(2^a - 1, 2^b - 1)$ are of the form $2^r - 1$, where r is a remainder arising when the Euclidean algorithm is used to find $\gcd(a, b)$.]
38. Use Exercise 37 to show that the integers $2^{35} - 1, 2^{34} - 1, 2^{33} - 1, 2^{31} - 1, 2^{29} - 1$, and $2^{23} - 1$ are pairwise relatively prime.
39. Using the method followed in Example 17, express the greatest common divisor of each of these pairs of integers as a linear combination of these integers.
 a) 10, 11 b) 21, 44 c) 36, 48
 d) 34, 55 e) 117, 213 f) 0, 223
 g) 123, 2347 h) 3454, 4666 i) 9999, 11111

40. Using the method followed in Example 17, express the greatest common divisor of each of these pairs of integers as a linear combination of these integers.
- a) 9, 11 b) 33, 44 c) 35, 78
 d) 21, 55 e) 101, 203 f) 124, 323
 g) 2002, 2339 h) 3457, 4669 i) 10001, 13422
41. Use the extended Euclidean algorithm to express $\gcd(26, 91)$ as a linear combination of 26 and 91.
42. Use the extended Euclidean algorithm to express $\gcd(252, 356)$ as a linear combination of 252 and 356.
43. Use the extended Euclidean algorithm to express $\gcd(144, 89)$ as a linear combination of 144 and 89.
44. Use the extended Euclidean algorithm to express $\gcd(1001, 100001)$ as a linear combination of 1001 and 100001.
45. Describe the extended Euclidean algorithm using pseudocode.
46. Find the smallest positive integer with exactly n different positive factors when n is
- a) 3. b) 4. c) 5.
 d) 6. e) 10.
47. Can you find a formula or rule for the n th term of a sequence related to the prime numbers or prime factorizations so that the initial terms of the sequence have these values?
- a) 0, 1, 1, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, ...
 b) 1, 2, 3, 2, 5, 2, 7, 2, 3, 2, 11, 2, 13, 2, ...
 c) 1, 2, 2, 3, 2, 4, 2, 4, 3, 4, 2, 6, 2, 4, ...
 d) 1, 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, ...
 e) 1, 2, 3, 3, 5, 5, 7, 7, 7, 11, 11, 13, 13, ...
 f) 1, 2, 6, 30, 210, 2310, 30030, 510510, 9699690, 223092870, ...
48. Can you find a formula or rule for the n th term of a sequence related to the prime numbers or prime factorizations so that the initial terms of the sequence have these values?
- a) 2, 2, 3, 5, 5, 7, 7, 11, 11, 11, 11, 13, 13, ...
 b) 0, 1, 2, 2, 3, 3, 4, 4, 4, 4, 5, 5, 6, 6, ...
 c) 1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, ...
 d) 1, -1, -1, 0, -1, 1, -1, 0, 0, 1, -1, 0, -1, 1, 1, ...
 e) 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 1, 0, 0, ...
 f) 4, 9, 25, 49, 121, 169, 289, 361, 529, 841, 961, 1369, ...
49. Prove that the product of any three consecutive integers is divisible by 6.
50. Show that if a, b , and m are integers such that $m \geq 2$ and $a \equiv b \pmod{m}$, then $\gcd(a, m) = \gcd(b, m)$.
- *51. Prove or disprove that $n^2 - 79n + 1601$ is prime whenever n is a positive integer.
52. Prove or disprove that $p_1 p_2 \cdots p_n + 1$ is prime for every positive integer n , where $p_1, p_2, \dots, p_n$ are the n smallest prime numbers.
53. Show that there is a composite integer in every arithmetic progression $ak + b$, $k = 1, 2, \dots$, where a and b are positive integers.
54. Adapt the proof in the text that there are infinitely many primes to prove that there are infinitely many primes of the form $3k + 2$, where k is a nonnegative integer. [Hint: Suppose that there are only finitely many such primes $q_1, q_2, \dots, q_n$, and consider the number $3q_1 q_2 \cdots q_n - 1$.]
55. Adapt the proof in the text that there are infinitely many primes to prove that there are infinitely many primes of the form $4k + 3$, where k is a nonnegative integer. [Hint: Suppose that there are only finitely many such primes $q_1, q_2, \dots, q_n$, and consider the number $4q_1 q_2 \cdots q_n - 1$.]
- *56. Prove that the set of positive rational numbers is countable by setting up a function that assigns to a rational number p/q with $\gcd(p, q) = 1$ the base 11 number formed by the decimal representation of p followed by the base 11 digit A, which corresponds to the decimal number 10, followed by the decimal representation of q .
- *57. Prove that the set of positive rational numbers is countable by showing that the function K is a one-to-one correspondence between the set of positive rational numbers and the set of positive integers if $K(m/n) = p_1^{2a_1} p_2^{2a_2} \cdots p_s^{2a_s} q_1^{2b_1-1} q_2^{2b_2-1} \cdots q_t^{2b_t-1}$, where $\gcd(m, n) = 1$ and the prime-power factorizations of m and n are $m = p_1^{a_1} p_2^{a_2} \cdots p_s^{a_s}$ and $n = q_1^{b_1} q_2^{b_2} \cdots q_t^{b_t}$.

4.4 Solving Congruences

4.4.1 Introduction

Solving linear congruences, which have the form $ax \equiv b \pmod{m}$, is an essential task in the study of number theory and its applications, just as solving linear equations plays an important role in calculus and linear algebra. To solve linear congruences, we employ inverses modulo m . We explain how to work backwards through the steps of the Euclidean algorithm to find inverses modulo m . Once we have found an inverse of a modulo m , we solve the congruence $ax \equiv b \pmod{m}$ by multiplying both sides of the congruence by this inverse.

Exercises

1. Show that 15 is an inverse of 7 modulo 26.
-  2. Show that 937 is an inverse of 13 modulo 2436.
3. By inspection (as discussed prior to Example 1), find an inverse of 4 modulo 9.
4. By inspection (as discussed prior to Example 1), find an inverse of 2 modulo 17.
5. Find an inverse of a modulo m for each of these pairs of relatively prime integers using the method followed in Example 2.
 - a) $a = 4, m = 9$
 - b) $a = 19, m = 141$
 - c) $a = 55, m = 89$
 - d) $a = 89, m = 232$
6. Find an inverse of a modulo m for each of these pairs of relatively prime integers using the method followed in Example 2.
 - a) $a = 2, m = 17$
 - b) $a = 34, m = 89$
 - c) $a = 144, m = 233$
 - d) $a = 200, m = 1001$
- *7. Show that if a and m are relatively prime positive integers, then the inverse of a modulo m is unique modulo m . [Hint: Assume that there are two solutions b and c of the congruence $ax \equiv 1 \pmod{m}$. Use Theorem 7 of Section 4.3 to show that $b \equiv c \pmod{m}$.]
8. Show that an inverse of a modulo m , where a is an integer and $m > 2$ is a positive integer, does not exist if $\gcd(a, m) > 1$.
9. Solve the congruence $4x \equiv 5 \pmod{9}$ using the inverse of 4 modulo 9 found in part (a) of Exercise 5.
10. Solve the congruence $2x \equiv 7 \pmod{17}$ using the inverse of 2 modulo 17 found in part (a) of Exercise 6.
11. Solve each of these congruences using the modular inverses found in parts (b), (c), and (d) of Exercise 5.
 - a) $19x \equiv 4 \pmod{141}$
 - b) $55x \equiv 34 \pmod{89}$
 - c) $89x \equiv 2 \pmod{232}$
12. Solve each of these congruences using the modular inverses found in parts (b), (c), and (d) of Exercise 6.
 - a) $34x \equiv 77 \pmod{89}$
 - b) $144x \equiv 4 \pmod{233}$
 - c) $200x \equiv 13 \pmod{1001}$
13. Find the solutions of the congruence $15x^2 + 19x \equiv 5 \pmod{11}$. [Hint: Show the congruence is equivalent to the congruence $15x^2 + 19x + 6 \equiv 0 \pmod{11}$. Factor the left-hand side of the congruence; show that a solution of the quadratic congruence is a solution of one of the two different linear congruences.]
14. Find the solutions of the congruence $12x^2 + 25x \equiv 10 \pmod{11}$. [Hint: Show the congruence is equivalent to the congruence $12x^2 + 25x + 12 \equiv 0 \pmod{11}$. Factor the left-hand side of the congruence; show that a solution of the quadratic congruence is a solution of one of two different linear congruences.]
- *15. Show that if m is an integer greater than 1 and $ac \equiv bc \pmod{m}$, then $a \equiv b \pmod{m/\gcd(c, m)}$.
16. a) Show that the positive integers less than 11, except 1 and 10, can be split into pairs of integers such that each pair consists of integers that are inverses of each other modulo 11.
b) Use part (a) to show that $10! \equiv -1 \pmod{11}$.
17. Show that if p is prime, the only solutions of $x^2 \equiv 1 \pmod{p}$ are integers x such that $x \equiv 1 \pmod{p}$ or $x \equiv -1 \pmod{p}$.
- *18. a) Generalize the result in part (a) of Exercise 16; that is, show that if p is a prime, the positive integers less than p , except 1 and $p - 1$, can be split into $(p - 3)/2$ pairs of integers such that each pair consists of integers that are inverses of each other. [Hint: Use the result of Exercise 17.]
b) From part (a) conclude that $(p - 1)! \equiv -1 \pmod{p}$ whenever p is prime. This result is known as **Wilson's theorem**.
c) What can we conclude if n is a positive integer such that $(n - 1)! \not\equiv -1 \pmod{n}$?
- *19. This exercise outlines a proof of Fermat's little theorem.
 - a) Suppose that a is not divisible by the prime p . Show that no two of the integers $1 \cdot a, 2 \cdot a, \dots, (p - 1)a$ are congruent modulo p .
 - b) Conclude from part (a) that the product of $1, 2, \dots, p - 1$ is congruent modulo p to the product of $a, 2a, \dots, (p - 1)a$. Use this to show that

$$(p - 1)! \equiv a^{p-1}(p - 1)! \pmod{p}.$$
 - c) Use Theorem 7 of Section 4.3 to show from part (b) that $a^{p-1} \equiv 1 \pmod{p}$ if $p \nmid a$. [Hint: Use Lemma 3 of Section 4.3 to show that p does not divide $(p - 1)!$ and then use Theorem 7 of Section 4.3. Alternatively, use Wilson's theorem from Exercise 18(b).]
 - d) Use part (c) to show that $a^p \equiv a \pmod{p}$ for all integers a .
20. Use the construction in the proof of the Chinese remainder theorem to find all solutions to the system of congruences $x \equiv 2 \pmod{3}$, $x \equiv 1 \pmod{4}$, and $x \equiv 3 \pmod{5}$.
21. Use the construction in the proof of the Chinese remainder theorem to find all solutions to the system of congruences $x \equiv 1 \pmod{2}$, $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, and $x \equiv 4 \pmod{11}$.
22. Solve the system of congruence $x \equiv 3 \pmod{6}$ and $x \equiv 4 \pmod{7}$ using the method of back substitution.
23. Solve the system of congruences in Exercise 20 using the method of back substitution.
24. Solve the system of congruences in Exercise 21 using the method of back substitution.

25. Write out in pseudocode an algorithm for solving a simultaneous system of linear congruences based on the construction in the proof of the Chinese remainder theorem.
- *26. Find all solutions, if any, to the system of congruences $x \equiv 5 \pmod{6}$, $x \equiv 3 \pmod{10}$, and $x \equiv 8 \pmod{15}$.
- *27. Find all solutions, if any, to the system of congruences $x \equiv 7 \pmod{9}$, $x \equiv 4 \pmod{12}$, and $x \equiv 16 \pmod{21}$.
28. Use the Chinese remainder theorem to show that an integer a , with $0 \leq a < m = m_1 m_2 \cdots m_n$, where the positive integers $m_1, m_2, \dots, m_n$ are pairwise relatively prime, can be represented uniquely by the n -tuple $(a \bmod m_1, a \bmod m_2, \dots, a \bmod m_n)$.
- *29. Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime integers greater than or equal to 2. Show that if $a \equiv b \pmod{m_i}$ for $i = 1, 2, \dots, n$, then $a \equiv b \pmod{m}$, where $m = m_1 m_2 \cdots m_n$. (This result will be used in Exercise 30 to prove the Chinese remainder theorem. Consequently, do not use the Chinese remainder theorem to prove it.)
- *30. Complete the proof of the Chinese remainder theorem by showing that the simultaneous solution of a system of linear congruences modulo pairwise relatively prime moduli is unique modulo the product of these moduli. [Hint: Assume that x and y are two simultaneous solutions. Show that $m_i \mid x - y$ for all i . Using Exercise 29, conclude that $m = m_1 m_2 \cdots m_n \mid x - y$.]
31. Which integers leave a remainder of 1 when divided by 2 and also leave a remainder of 1 when divided by 3?
32. Which integers are divisible by 5 but leave a remainder of 1 when divided by 3?
33. Use Fermat's little theorem to find $7^{121} \bmod 13$.
34. Use Fermat's little theorem to find $23^{1002} \bmod 41$.
35. Use Fermat's little theorem to show that if p is prime and $p \nmid a$, then a^{p-2} is an inverse of a modulo p .
36. Use Exercise 35 to find an inverse of 5 modulo 41.
37. a) Show that $2^{340} \equiv 1 \pmod{11}$ by Fermat's little theorem and noting that $2^{340} = (2^{10})^{34}$.
b) Show that $2^{340} \equiv 1 \pmod{31}$ using the fact that $2^{340} = (2^5)^{68} = 32^{68}$.
c) Conclude from parts (a) and (b) that $2^{340} \equiv 1 \pmod{341}$.
38. a) Use Fermat's little theorem to compute $3^{302} \bmod 5$, $3^{302} \bmod 7$, and $3^{302} \bmod 11$.
b) Use your results from part (a) and the Chinese remainder theorem to find $3^{302} \bmod 385$. (Note that $385 = 5 \cdot 7 \cdot 11$.)
39. a) Use Fermat's little theorem to compute $5^{2003} \bmod 7$, $5^{2003} \bmod 11$, and $5^{2003} \bmod 13$.
b) Use your results from part (a) and the Chinese remainder theorem to find $5^{2003} \bmod 1001$. (Note that $1001 = 7 \cdot 11 \cdot 13$.)
40. Show with the help of Fermat's little theorem that if n is a positive integer, then 42 divides $n^7 - n$.
41. Show that if p is an odd prime, then every divisor of the Mersenne number $2^p - 1$ is of the form $2kp + 1$, where k is a nonnegative integer. [Hint: Use Fermat's little theorem and Exercise 37 of Section 4.3.]

42. Use Exercise 41 to determine whether $M_{13} = 2^{13} - 1 = 8191$ and $M_{23} = 2^{23} - 1 = 8,388,607$ are prime.
43. Use Exercise 41 to determine whether $M_{11} = 2^{11} - 1 = 2047$ and $M_{17} = 2^{17} - 1 = 131,071$ are prime.

✎ Let n be a positive integer and let $n - 1 = 2^s t$, where s is a nonnegative integer and t is an odd positive integer. We say that n passes **Miller's test for the base b** if either $b^t \equiv 1 \pmod{n}$ or $b^{2^j t} \equiv -1 \pmod{n}$ for some j with $0 \leq j < s - 1$. It can be shown (see [Ro10]) that a composite integer n passes Miller's test for fewer than $n/4$ bases b with $1 < b < n$. A composite positive integer n that passes Miller's test to the base b is called a **strong pseudoprime to the base b** .

- *44. Show that if n is prime and b is a positive integer with $n \nmid b$, then n passes Miller's test to the base b .
45. Show that 2047 is a strong pseudoprime to the base 2 by showing that it passes Miller's test to the base 2, but is composite.
46. Show that 1729 is a Carmichael number.
47. Show that 2821 is a Carmichael number.
- *48. Show that if $n = p_1 p_2 \cdots p_k$, where $p_1, p_2, \dots, p_k$ are distinct primes that satisfy $p_j - 1 \mid n - 1$ for $j = 1, 2, \dots, k$, then n is a Carmichael number.
49. a) Use Exercise 48 to show that every integer of the form $(6m + 1)(12m + 1)(18m + 1)$, where m is a positive integer and $6m + 1$, $12m + 1$, and $18m + 1$ are all primes, is a Carmichael number.
b) Use part (a) to show that 172,947,529 is a Carmichael number.
50. Find the nonnegative integer a less than 28 represented by each of these pairs, where each pair represents $(a \bmod 4, a \bmod 7)$.
- | | | |
|-----------|-----------|-----------|
| a) (0, 0) | b) (1, 0) | c) (1, 1) |
| d) (2, 1) | e) (2, 2) | f) (0, 3) |
| g) (2, 0) | h) (3, 5) | i) (3, 6) |
51. Express each nonnegative integer a less than 15 as a pair $(a \bmod 3, a \bmod 5)$.
52. Explain how to use the pairs found in Exercise 51 to add 4 and 7.
53. Solve the system of congruences that arises in Example 8.
54. Show that 2 is a primitive root of 19.
55. Find the discrete logarithms of 5 and 6 to the base 2 modulo 19.
56. Let p be an odd prime and r a primitive root of p . Show that if a and b are positive integers in $\mathbb{Z}_p$, then $\log_r(ab) \equiv \log_r a + \log_r b \pmod{p - 1}$.
57. Write out a table of discrete logarithms modulo 17 with respect to the primitive root 3.
- If m is a positive integer, the integer a is a **quadratic residue** of m if $\gcd(a, m) = 1$ and the congruence $x^2 \equiv a \pmod{m}$ has a solution. In other words, a quadratic residue of m is an integer relatively prime to m that is a perfect square modulo m . If a is not a quadratic residue of m and $\gcd(a, m) = 1$, we say that it is a **quadratic nonresidue** of m . For example, 2 is a quadratic residue of 7 because $\gcd(2, 7) = 1$ and $3^2 \equiv 2 \pmod{7}$ and 3 is a quadratic nonresidue of 7 because $\gcd(3, 7) = 1$ and $x^2 \equiv 3 \pmod{7}$ has no solution.

58. Which integers are quadratic residues of 11?
59. Show that if p is an odd prime and a is an integer not divisible by p , then the congruence $x^2 \equiv a \pmod{p}$ has either no solutions or exactly two incongruent solutions modulo p .

60. Show that if p is an odd prime, then there are exactly $(p-1)/2$ quadratic residues of p among the integers $1, 2, \dots, p-1$.

If p is an odd prime and a is an integer not divisible by p , the **Legendre symbol** $\left(\frac{a}{p}\right)$ is defined to be 1 if a is a quadratic residue of p and -1 otherwise.

61. Show that if p is an odd prime and a and b are integers with $a \equiv b \pmod{p}$, then

$$\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right).$$

62. Prove **Euler's criterion**, which states that if p is an odd prime and a is a positive integer not divisible by p , then

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

[Hint: If a is a quadratic residue modulo p , apply Fermat's little theorem; otherwise, apply Wilson's theorem, given in Exercise 18(b).]

63. Use Exercise 62 to show that if p is an odd prime and a and b are integers not divisible by p , then

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right).$$

64. Show that if p is an odd prime, then -1 is a quadratic residue of p if $p \equiv 1 \pmod{4}$, and -1 is not a quadratic residue of p if $p \equiv 3 \pmod{4}$. [Hint: Use Exercise 62.]
65. Find all solutions of the congruence $x^2 \equiv 29 \pmod{35}$. [Hint: Find the solutions of this congruence modulo 5 and modulo 7, and then use the Chinese remainder theorem.]
66. Find all solutions of the congruence $x^2 \equiv 16 \pmod{105}$. [Hint: Find the solutions of this congruence modulo 3, modulo 5, and modulo 7, and then use the Chinese remainder theorem.]
67. Describe a brute-force algorithm for solving the discrete logarithm problem and find the worst-case and average-case time complexity of this algorithm.

4.5 Applications of Congruences

Congruences have many applications to discrete mathematics, computer science, and many other disciplines. We will introduce three applications in this section: the use of congruences to assign memory locations to computer files, the generation of pseudorandom numbers, and check digits.

Suppose that a customer identification number is ten digits long. To retrieve customer files quickly, we do not want to assign a memory location to a customer record using the ten-digit identification number. Instead, we want to use a smaller integer associated to the identification number. This can be done using what is known as a hashing function. In this section we will show how we can use modular arithmetic to do hashing.

Constructing sequences of random numbers is important for randomized algorithms, for simulations, and for many other purposes. Constructing a sequence of truly random numbers is extremely difficult, or perhaps impossible, because any method for generating what are supposed to be random numbers may generate numbers with hidden patterns. As a consequence, methods have been developed for finding sequences of numbers that have many desirable properties of random numbers, and which can be used for various purposes in place of random numbers. In this section we will show how to use congruences to generate sequences of pseudorandom numbers. The advantage is that the pseudorandom numbers so generated are constructed quickly; the disadvantage is that they have too much predictability to be used for many tasks.

Congruences also can be used to produce check digits for identification numbers of various kinds, such as code numbers used to identify retail products, numbers used to identify books, airline ticket numbers, and so on. We will explain how to construct check digits using congruences for a variety of types of identification numbers. We will show that these check digits can be used to detect certain kinds of common errors made when identification numbers are printed.

Publishers sometimes do not calculate ISBNs correctly for their books, as was done for an earlier edition of this text.

Several kinds of errors often arise in identification numbers. A **single error**, an error in one digit of an identification number, is perhaps the most common type of error. Another common kind of error is a **transposition error**, which occurs when two digits are accidentally interchanged. For each type of identification number, including a check digit, we would like to be able to detect these common types of errors, as well as other types of errors. We will investigate whether the check digit for ISBNs can detect single errors and transposition errors. Whether check digits for UPCs can detect these kinds of errors is left as Exercises 26 and 27.

Suppose that $x_1x_2 \dots x_{10}$ is a valid ISBN (so that $\sum_{i=1}^{10} x_i \equiv 0 \pmod{10}$). We will show that we can detect a single error and a transposition of two digits (where we include the possibility that one of the two digits is the check digit X, representing 10). Suppose that this ISBN has been printed with a single error as $y_1y_2 \dots y_{10}$. If there is a single error, then, for some integer j , $y_i = x_i$ for $i \neq j$ and $y_j = x_j + a$, where $-10 \leq a \leq 10$ and $a \neq 0$. Note that $a = y_j - x_j$ is the error in the j th place. It then follows that

$$\sum_{i=1}^{10} iy_i = \left(\sum_{i=1}^{10} ix_i \right) + ja \equiv ja \not\equiv 0 \pmod{11}.$$

These last two congruences hold because $\sum_{i=1}^{10} x_i \equiv 0 \pmod{10}$ and $11 \nmid ja$, because $11 \nmid j$ and $11 \nmid a$. We conclude that $y_1y_2 \dots y_{10}$ is not a valid ISBN. So, we have detected the single error.

Now suppose that two unequal digits have been transposed. It follows that there are distinct integers j and k such that $y_j = x_k$ and $y_k = x_j$, and $y_i = x_i$ for $i \neq j$ and $i \neq k$. Hence,

$$\sum_{i=1}^{10} iy_i = \left(\sum_{i=1}^{10} ix_i \right) + (jx_k - jx_j) + (kx_j - kx_k) \equiv (j - k)(x_k - x_j) \not\equiv 0 \pmod{11},$$

because $\sum_{i=1}^{10} x_i \equiv 0 \pmod{10}$ and $11 \nmid (j - k)$ and $11 \nmid (x_k - x_j)$. We see that $y_1y_2 \dots y_{10}$ is not a valid ISBN. Thus, we can detect the interchange of two unequal digits.

Exercises

- Which memory locations are assigned by the hashing function $h(k) = k \bmod 97$ to the records of insurance company customers with these Social Security numbers?
 - 034567981
 - 183211232
 - 220195744
 - 987255335
- Which memory locations are assigned by the hashing function $h(k) = k \bmod 101$ to the records of insurance company customers with these Social Security numbers?
 - 104578690
 - 432222187
 - 372201919
 - 501338753
- A parking lot has 31 visitor spaces, numbered from 0 to 30. Visitors are assigned parking spaces using the hashing function $h(k) = k \bmod 31$, where k is the number formed from the first three digits on a visitor's license plate.
 - Which spaces are assigned by the hashing function to cars that have these first three digits on their license plates: 317, 918, 007, 100, 111, 310?
 - Describe a procedure visitors should follow to find a free parking space, when the space they are assigned is occupied.

Another way to resolve collisions in hashing is to use *double hashing*. We use an initial hashing function $h(k) = k \bmod p$, where p is prime. We also use a second hashing function $g(k) = (k + 1) \bmod (p - 2)$. When a collision occurs, we use a *probing sequence* $h(k, i) = (h(k) + i \cdot g(k)) \bmod p$.

- Use the double hashing procedure we have described with $p = 4969$ to assign memory locations to files for employees with social security numbers $k_1 = 132489971$, $k_2 = 509496993$, $k_3 = 546332190$, $k_4 = 034367980$, $k_5 = 047900151$, $k_6 = 329938157$, $k_7 = 212228844$, $k_8 = 325510778$, $k_9 = 353354519$, $k_{10} = 053708912$.
- What sequence of pseudorandom numbers is generated using the linear congruential generator $x_{n+1} = (3x_n + 2) \bmod 13$ with seed $x_0 = 1$?
- What sequence of pseudorandom numbers is generated using the linear congruential generator $x_{n+1} = (4x_n + 1) \bmod 7$ with seed $x_0 = 3$?
- What sequence of pseudorandom numbers is generated using the pure multiplicative generator $x_{n+1} = 3x_n \bmod 11$ with seed $x_0 = 2$?

8. Write an algorithm in pseudocode for generating a sequence of pseudorandom numbers using a linear congruential generator.

The **middle-square method** for generating pseudorandom numbers begins with an n -digit integer. This number is squared, initial zeros are appended to ensure that the result has $2n$ digits, and its middle n digits are used to form the next number in the sequence. This process is repeated to generate additional terms.

9. Find the first eight terms of the sequence of four-digit pseudorandom numbers generated by the middle square method starting with 2357.
10. Explain why both 3792 and 2916 would be bad choices for the initial term of a sequence of four-digit pseudorandom numbers generated by the middle square method.
- The **power generator** is a method for generating pseudorandom numbers. To use the power generator, parameters p and d are specified, where p is a prime, d is a positive integer such that $p \nmid d$, and a seed x_0 is specified. The pseudorandom numbers $x_1, x_2, \dots$ are generated using the recursive definition $x_{n+1} = x_n^d \bmod p$.
11. Find the sequence of pseudorandom numbers generated by the power generator with $p = 7$, $d = 3$, and seed $x_0 = 2$.
12. Find the sequence of pseudorandom numbers generated by the power generator with $p = 11$, $d = 2$, and seed $x_0 = 3$.
13. Suppose you received these bit strings over a communications link, where the last bit is a parity check bit. In which string are you sure there is an error?
- a) 00000111111 b) 10101010101
c) 11111100000 d) 10111101111
14. Prove that a parity check bit can detect an error in a string if and only if the string contains an odd number of errors.

15. The first nine digits of the ISBN-10 of the European version of the fifth edition of this book are 0-07-119881. What is the check digit for that book?
16. The ISBN-10 of the sixth edition of *Elementary Number Theory and Its Applications* is 0-321-500Q1-8, where Q is a digit. Find the value of Q .
17. Determine whether the check digit of the ISBN-10 for this textbook (the eighth edition of *Discrete Mathematics and Its Applications*) was computed correctly by the publisher.

The United States Postal Service (USPS) sells money orders identified by an 11-digit number $x_1x_2 \dots x_{11}$. The first ten digits identify the money order; x_{11} is a check digit that satisfies $x_{11} = x_1 + x_2 + \dots + x_{10} \bmod 9$.

18. Find the check digit for the USPS money orders that have identification number that start with these ten digits.
- a) 7555618873 b) 6966133421
c) 8018927435 d) 3289744134
19. Determine whether each of these numbers is a valid USPS money order identification number.
- a) 74051489623 b) 88382013445
c) 56152240784 d) 66606631178
20. One digit in each of these identification numbers of a postal money order is smudged. Can you recover the smudged digit, indicated by a Q , in each of these numbers?
- a) $Q1223139784$ b) $6702120Q988$
c) $27Q41007734$ d) $213279032Q1$
21. One digit in each of these identification numbers of a postal money order is smudged. Can you recover the smudged digit, indicated by a Q , in each of these numbers?
- a) $493212Q0688$ b) $850Q9103858$
c) $2Q941007734$ d) $66687Q03201$
22. Determine which single digit errors are detected by the USPS money order code.
23. Determine which transposition errors are detected by the USPS money order code.
24. Determine the check digit for the UPCs that have these initial 11 digits.
- a) 73232184434 b) 63623991346
c) 04587320720 d) 93764323341
25. Determine whether each of the strings of 12 digits is a valid UPC code.
- a) 036000291452 b) 012345678903
c) 782421843014 d) 726412175425
26. Does the check digit of a UPC code detect all single errors? Prove your answer or find a counterexample.
27. Determine which transposition errors the check digit of a UPC code finds.

Some airline tickets have a 15-digit identification number $a_1a_2 \dots a_{15}$, where a_{15} is a check digit that equals $a_1a_2 \dots a_{14} \bmod 7$.

28. Find the check digit a_{15} that follows each of these initial 14 digits of an airline ticket identification number.
- a) 10237424413392 b) 00032781811234
c) 00611232134231 d) 00193222543435
29. Determine whether each of these 15-digit numbers is a valid airline ticket identification number.
- a) 101333341789013 b) 007862342770445
c) 113273438882531 d) 000122347322871
30. Which errors in a single digit of a 15-digit airline ticket identification number can be detected?
- *31. Can the accidental transposition of two consecutive digits in an airline ticket identification number be detected using the check digit?

Periodicals are identified using an **International Standard Serial Number (ISSN)**. An ISSN consists of two blocks of four digits. The last digit in the second block is a check digit. This check digit is determined by the congruence $d_8 \equiv 3d_1 + 4d_2 + 5d_3 + 6d_4 + 7d_5 + 8d_6 + 9d_7 \pmod{11}$. When $d_8 \equiv 10 \pmod{11}$, we use the letter X to represent d_8 in the code.

32. For each of these initial seven digits of an ISSN, determine the check digit (which may be the letter X).
- | | |
|-------------|-------------|
| a) 1570-868 | b) 1553-734 |
| c) 1089-708 | d) 1383-811 |
33. Are each of these eight-digit codes possible ISSNs? That is, do they end with a correct check digit?
- | | |
|--------------|--------------|
| a) 1059-1027 | b) 0002-9890 |
| c) 1530-8669 | d) 1007-120X |
34. Does the check digit of an ISSN detect every single error in an ISSN? Justify your answer with either a proof or a counterexample.
35. Does the check digit of an ISSN detect every error where two consecutive digits are accidentally interchanged? Justify your answer with either a proof or a counterexample.

4.6 Cryptography

4.6.1 Introduction

Number theory plays a key role in cryptography, the subject of transforming information so that it cannot be easily recovered without special knowledge. Number theory is the basis of many classical ciphers, first used thousands of years ago, and used extensively until the 20th century. These ciphers encrypt messages by changing each letter to a different letter, or each block of letters to a different block of letters. We will discuss some classical ciphers, including shift ciphers, which replace each letter by the letter a fixed number of positions later in the alphabet, wrapping around to the beginning of the alphabet when necessary. The classical ciphers we will discuss are examples of private key ciphers where knowing how to encrypt allows someone to also decrypt messages. With a private key cipher, two parties who wish to communicate in secret must share a secret key. The classical ciphers we will discuss are also vulnerable to cryptanalysis, which seeks to recover encrypted information without access to the secret information used to encrypt the message. We will show how to cryptanalyze messages sent using shift ciphers.

Number theory is also important in public key cryptography, a type of cryptography invented in the 1970s. In public key cryptography, knowing how to encrypt does not also tell someone how to decrypt. The most widely used public key system, called the RSA cryptosystem, encrypts messages using modular exponentiation, where the modulus is the product of two large primes. Knowing how to encrypt requires that someone know the modulus and an exponent. (It does not require that the two prime factors of the modulus be known.) As far as it is known, knowing how to decrypt requires someone to know how to invert the encryption function, which can only be done in a practical amount of time when someone knows these two large prime factors. In this chapter we will explain how the RSA cryptosystem works, including how to encrypt and decrypt messages.

The subject of cryptography also includes the subject of cryptographic protocols, which are exchanges of messages carried out by two or more parties to achieve a specific security goal. We will discuss two important protocols in this chapter. One allows two people to share a common secret key. The other can be used to send signed messages so that a recipient can be sure that they were sent by the purported sender. Finally, we will introduce the notion of a homomorphic cryptosystem, now playing an important role in cloud computing. Data can become vulnerable if they must be decrypted for use as input to programs. Homomorphic encryption eliminates this vulnerability by allowing programs to be run on encrypted data. The output of these programs is then the encryption of the desired output.

Exercises

1. Encrypt the message DO NOT PASS GO by translating the letters into numbers, applying the given encryption function, and then translating the numbers back into letters.
 - a) $f(p) = (p + 3) \bmod 26$ (the Caesar cipher)
 - b) $f(p) = (p + 13) \bmod 26$
 - c) $f(p) = (3p + 7) \bmod 26$
 2. Encrypt the message STOP POLLUTION by translating the letters into numbers, applying the given encryption function, and then translating the numbers back into letters.
 - a) $f(p) = (p + 4) \bmod 26$
 - b) $f(p) = (p + 21) \bmod 26$
 - c) $f(p) = (17p + 22) \bmod 26$
 3. Encrypt the message WATCH YOUR STEP by translating the letters into numbers, applying the given encryption function, and then translating the numbers back into letters.
 - a) $f(p) = (p + 14) \bmod 26$
 - b) $f(p) = (14p + 21) \bmod 26$
 - c) $f(p) = (-7p + 1) \bmod 26$
 4. Decrypt these messages that were encrypted using the Caesar cipher.
 - a) EOXH MHDQV
 - b) WHVW WRGDB
 - c) HDW GLP VXP
 5. Decrypt these messages encrypted using the shift cipher $f(p) = (p + 10) \bmod 26$.
 - a) CEBBOXNOB XYG
 - b) LO WI PBSOXN
 - c) DSWO PYB PEX
 6. Suppose that when a long string of text is encrypted using a shift cipher $f(p) = (p + k) \bmod 26$, the most common letter in the ciphertext is X. What is the most likely value for k , assuming that the distribution of letters in the text is typical of English text?
 7. Suppose that when a string of English text is encrypted using a shift cipher $f(p) = (p + k) \bmod 26$, the resulting ciphertext is DY CVOOZ ZOBMRKXMO DY NBOKW. What was the original plaintext string?
 8. Suppose that the ciphertext DVE CFMV KF NFEUVI, REU KYRK ZJ KYV JVVU FW JTZVETV was produced by encrypting a plaintext message using a shift cipher. What is the original plaintext?
 9. Suppose that the ciphertext ERC WYJMG MIRXPC EHZERGIH XIGLRSPSKC MW MRHMXM-RKYMWLEFPI JVSQ QEKMG was produced by encrypting a plaintext message using a shift cipher. What is the original plaintext?
 10. Determine whether there is a key for which the enciphering function for the shift cipher is the same as the deciphering function.
 11. What is the decryption function for an affine cipher if the encryption function is $c = (15p + 13) \bmod 26$?
 - *12. Find all pairs of integers keys (a, b) for affine ciphers for which the encryption function $c = (ap + b) \bmod 26$ is the same as the corresponding decryption function.
 13. Suppose that the most common letter and the second most common letter in a long ciphertext produced by encrypting a plaintext using an affine cipher $f(p) = (ap + b) \bmod 26$ are Z and J, respectively. What are the most likely values of a and b ?
 14. Encrypt the message GRIZZLY BEARS using blocks of five letters and the transposition cipher based on the permutation of $\{1, 2, 3, 4, 5\}$ with $\sigma(1) = 3$, $\sigma(2) = 5$, $\sigma(3) = 1$, $\sigma(4) = 2$, and $\sigma(5) = 4$. For this exercise, use the letter X as many times as necessary to fill out the final block of fewer than five letters.
 15. Decrypt the message EABW EFRO ATMR ASIN, which is the ciphertext produced by encrypting a plaintext message using the transposition cipher with blocks of four letters and the permutation σ of $\{1, 2, 3, 4\}$ defined by $\sigma(1) = 3$, $\sigma(2) = 1$, $\sigma(3) = 4$, and $\sigma(4) = 2$.
 - *16. Suppose that you know that a ciphertext was produced by encrypting a plaintext message with a transposition cipher. How might you go about breaking it?
 17. Suppose you have intercepted a ciphertext message and when you determine the frequencies of letters in this message, you find the frequencies are similar to the frequency of letters in English text. Which type of cipher do you suspect was used?
- The **Vigenère cipher** is a block cipher, with a key that is a string of letters with numerical equivalents $k_1 k_2 \dots k_m$, where $k_i \in \mathbb{Z}_{26}$ for $i = 1, 2, \dots, m$. Suppose that the numerical equivalents of the letters of a plaintext block are $p_1 p_2 \dots p_m$. The corresponding numerical ciphertext block is $(p_1 + k_1) \bmod 26$ $(p_2 + k_2) \bmod 26 \dots (p_m + k_m) \bmod 26$. Finally, we translate back to letters. For example, suppose that the key string is RED, with numerical equivalents 17 4 3. Then, the plaintext ORANGE, with numerical equivalents 14 17 00 13 06 04, is encrypted by first splitting it into two blocks 14 17 00 and 13 06 04. Then, in each block we shift the first letter by 17, the second by 4, and the third by 3. We obtain 5 21 03 and 04 10 07. The ciphertext is FVDEKH.
18. Use the Vigenère cipher with key BLUE to encrypt the message SNOWFALL.
 19. The ciphertext OIKYWVHBX was produced by encrypting a plaintext message using the Vigenère cipher with key HOT. What is the plaintext message?
 20. Express the Vigenère cipher as a cryptosystem.
- To break a Vigenère cipher by recovering a plaintext message from the ciphertext message without having the key, the first step is to figure out the length of the key string. The second

step is to figure out each character of the key string by determining the corresponding shift. Exercises 21 and 22 deal with these two aspects.

21. Suppose that when a long string of text is encrypted using a Vigenère cipher, the same string is found in the ciphertext starting at several different positions. Explain how this information can be used to help determine the length of the key.
22. Once the length of the key string of a Vigenère cipher is known, explain how to determine each of its characters. Assume that the plaintext is long enough so that the frequency of its letters is reasonably close to the frequency of letters in typical English text.
- *23. Show that we can easily factor n when we know that n is the product of two primes, p and q , and we know the value of $(p-1)(q-1)$.

In Exercises 24–27 first express your answers without computing modular exponentiations. Then use a computational aid to complete these computations.

24. Encrypt the message ATTACK using the RSA system with $n = 43 \cdot 59$ and $e = 13$, translating each letter into integers and grouping together pairs of integers, as done in Example 8.
25. Encrypt the message UPLOAD using the RSA system with $n = 53 \cdot 61$ and $e = 17$, translating each letter into integers and grouping together pairs of integers, as done in Example 8.
26. What is the original message encrypted using the RSA system with $n = 53 \cdot 61$ and $e = 17$ if the encrypted message is 3185 2038 2460 2550? (To decrypt, first find the decryption exponent d , which is the inverse of $e = 17$ modulo $52 \cdot 60$.)
27. What is the original message encrypted using the RSA system with $n = 43 \cdot 59$ and $e = 13$ if the encrypted message is 0667 1947 0671? (To decrypt, first find the decryption exponent d which is the inverse of $e = 13$ modulo $42 \cdot 58$.)
- *28. Suppose that (n, e) is an RSA encryption key, with $n = pq$, where p and q are large primes and $\gcd(e, (p-1)(q-1)) = 1$. Furthermore, suppose that d is an inverse of e modulo $(p-1)(q-1)$. Suppose that $C \equiv M^e \pmod{pq}$. In the text we showed that RSA decryption, that is, the congruence $C^d \equiv M \pmod{pq}$ holds when $\gcd(M, pq) = 1$. Show that this decryption congruence also holds when $\gcd(M, pq) > 1$. [Hint: Use congruences modulo p and modulo q and apply the Chinese remainder theorem.]
29. Describe the steps that Alice and Bob follow when they use the Diffie-Hellman key exchange protocol to generate a shared key. Assume that they use the prime $p = 23$ and take $a = 5$, which is a primitive root of 23, and that Alice selects $k_1 = 8$ and Bob selects $k_2 = 5$. (You may want to use some computational aid.)
30. Describe the steps that Alice and Bob follow when they use the Diffie-Hellman key exchange protocol to generate a shared key. Assume that they use the prime $p = 101$ and take $a = 2$, which is a primitive root of 101, and that

Alice selects $k_1 = 7$ and Bob selects $k_2 = 9$. (You may want to use some computational aid.)

In Exercises 31–32 suppose that Alice and Bob have these public keys and corresponding private keys: $(n_{\text{Alice}}, e_{\text{Alice}}) = (2867, 7) = (61 \cdot 47, 7)$, $d_{\text{Alice}} = 1183$ and $(n_{\text{Bob}}, e_{\text{Bob}}) = (3127, 21) = (59 \cdot 53, 21)$, $d_{\text{Bob}} = 1149$. First express your answers without carrying out the calculations. Then, using a computational aid, if available, perform the calculation to get the numerical answers.

31. Alice wants to send to all her friends, including Bob, the message “SELL EVERYTHING” so that he knows that she sent it. What should she send to her friends, assuming she signs the message using the RSA cryptosystem.
32. Alice wants to send to Bob the message “BUY NOW” so that he knows that she sent it and so that only Bob can read it. What should she send to Bob, assuming she signs the message and then encrypts it using Bob’s public key?
33. We describe a basic key exchange protocol using private key cryptography upon which more sophisticated protocols for key exchange are based. Encryption within the protocol is done using a private key cryptosystem (such as AES) that is considered secure. The protocol involves three parties, Alice and Bob, who wish to exchange a key, and a trusted third party Cathy. Assume that Alice has a secret key k_{Alice} that only she and Cathy know, and Bob has a secret key k_{Bob} which only he and Cathy know. The protocol has three steps:

(i) Alice sends the trusted third party Cathy the message “request a shared key with Bob” encrypted using Alice’s key k_{Alice} .

(ii) Cathy sends back to Alice a key $k_{\text{Alice}, \text{Bob}}$, which she generates, encrypted using the key k_{Alice} , followed by this same key $k_{\text{Alice}, \text{Bob}}$ encrypted using Bob’s key, k_{Bob} .

(iii) Alice sends to Bob the key $k_{\text{Alice}, \text{Bob}}$ encrypted using k_{Bob} , known only to Bob and to Cathy.

Explain why this protocol allows Alice and Bob to share the secret key $k_{\text{Alice}, \text{Bob}}$, known only to them and to Cathy.

The **Paillier cryptosystem** is a public key cryptosystem described in 1999 by P. Paillier, used in some electronic voting systems. A public key (n, g) and a corresponding private key (p, q) are created by randomly selecting primes p and q so that $\gcd(pq, \lambda) = 1$, where $\lambda = (p-1)(q-1)$, and a nonzero element g of $\mathbf{Z}_{n^2}$ with $\gcd((g^\lambda \bmod n^2 - 1)/n, n) = 1$. To encrypt a message $m \in \mathbf{Z}_n$ a random nonzero element r of $\mathbf{Z}_n$ is first selected, and then used to find $c = g^m r^n \bmod n^2$.

34. a) Show that we can take $p = 149$, $q = 179$, and $g = 5$ to generate a public key for the Paillier cryptosystem by checking that all the conditions required for these parameters hold, and find the public and private keys that are generated.
b) Find the ciphertext corresponding to the plaintext $m = 67$ where we choose $r = 81$.
35. Show that the Paillier cryptosystem is additively homomorphic.

Key Terms and Results

TERMS

$a \mid b$ (a divides b): there is an integer c such that $b = ac$

a and b are congruent modulo m : m divides $a - b$

modular arithmetic: arithmetic done modulo an integer $m \geq 2$

prime: an integer greater than 1 with exactly two positive integer divisors

composite: an integer greater than 1 that is not prime

Mersenne prime: a prime of the form $2^p - 1$, where p is prime

$\gcd(a, b)$ (greatest common divisor of a and b): the largest integer that divides both a and b

relatively prime integers: integers a and b such that $\gcd(a, b) = 1$

pairwise relatively prime integers: a set of integers with the property that every pair of these integers is relatively prime

$\text{lcm}(a, b)$ (least common multiple of a and b): the smallest positive integer that is divisible by both a and b

$a \bmod b$: the remainder when the integer a is divided by the positive integer b

$a \equiv b \pmod{m}$ (a is congruent to b modulo m): $a - b$ is divisible by m

$n = (a_k a_{k-1} \dots a_1 a_0)_b$: the base b representation of n

binary representation: the base 2 representation of an integer

octal representation: the base 8 representation of an integer

hexadecimal representation: the base 16 representation of an integer

linear combination of a and b with integer coefficients: an expression of the form $sa + tb$, where s and t are integers

Bézout coefficients of a and b : integers s and t such that the Bézout identity $sa + tb = \gcd(a, b)$ holds

inverse of a modulo m : an integer $\bar{a}$ such that $\bar{a}a \equiv 1 \pmod{m}$

linear congruence: a congruence of the form $ax \equiv b \pmod{m}$, where x is an integer variable

pseudoprime to the base b : a composite integer n such that $b^{n-1} \equiv 1 \pmod{n}$

Carmichael number: a composite integer n such that n is a pseudoprime to the base b for all positive integers b with $\gcd(b, n) = 1$

primitive root of a prime p : an integer r in $\mathbf{Z}_p$ such that every integer not divisible by p is congruent modulo p to a power of r

discrete logarithm of a to the base r modulo p : the integer e with $0 \leq e \leq p - 1$ such that $r^e \equiv a \pmod{p}$

encryption: the process of making a message secret

decryption: the process of returning a secret message to its original form

encryption key: a value that determines which of a family of encryption functions is to be used

shift cipher: a cipher that encrypts the plaintext letter p as $(p + k) \bmod m$ for an integer k

affine cipher: a cipher that encrypts the plaintext letter p as $(ap + b) \bmod m$ for integers a and b with $\gcd(a, m) = 1$

character cipher: a cipher that encrypts characters one by one

block cipher: a cipher that encrypts blocks of characters of a fixed size

cryptanalysis: the process of recovering the plaintext from ciphertext without knowledge of the encryption method, or with knowledge of the encryption method, but not the key

cryptosystem: a five-tuple $(\mathcal{P}, \mathcal{C}, \mathcal{K}, \mathcal{E}, \mathcal{D})$ where $\mathcal{P}$ is the set of plaintext messages, $\mathcal{C}$ is the set of ciphertext messages, $\mathcal{K}$ is the set of keys, $\mathcal{E}$ is the set of encryption functions, and $\mathcal{D}$ is the set of decryption functions

private key encryption: encryption where both encryption keys and decryption keys must be kept secret

public key encryption: encryption where encryption keys are public knowledge, but decryption keys are kept secret

RSA cryptosystem: the cryptosystem where $\mathcal{P}$ and $\mathcal{C}$ are both $\mathbf{Z}_{26}$, $\mathcal{K}$ is the set of pairs $k = (n, e)$ where $n = pq$ where p and q are large primes and e is a positive integer, $E_k(p) = p^e \bmod n$, and $D_k(c) = c^d \bmod n$ where d is the inverse of e modulo $(p - 1)(q - 1)$

key exchange protocol: a protocol used for two parties to generate a shared key

digital signature: a method that a recipient can use to determine that the purported sender of a message actually sent the message

fully homomorphic cryptosystem: a cryptosystem that allows arbitrary computations to be run on encrypted data so that the output is the encryption of the unencrypted output of the unencrypted input

RESULTS

division algorithm: Let a and d be integers with d positive. Then there are unique integers q and r with $0 \leq r < d$ such that $a = dq + r$.

Let b be an integer greater than 1. Then if n is a positive integer, it can be expressed uniquely in the form $n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$.

The algorithm for finding the base b expansion of an integer (see Algorithm 1 in Section 4.2)

The conventional algorithms for addition and multiplication of integers (given in Section 4.2)

The fast modular exponentiation algorithm (see Algorithm 5 in Section 4.2)

Euclidean algorithm: for finding greatest common divisors by successively using the division algorithm (see Algorithm 1 in Section 4.3)

Bézout's theorem: If a and b are positive integers, then $\gcd(a, b)$ is a linear combination of a and b .

sieve of Eratosthenes: a procedure for finding all primes not exceeding a specified number n , described in Section 4.3

fundamental theorem of arithmetic: Every positive integer can be written uniquely as the product of primes, where the prime factors are written in order of increasing size.

If a and b are positive integers, then $ab = \gcd(a, b) \cdot \text{lcm}(a, b)$.

If m is a positive integer and $\gcd(a, m) = 1$, then a has a unique inverse modulo m .

Chinese remainder theorem: A system of linear congruences modulo pairwise relatively prime integers has a unique solution modulo the product of these moduli.

Fermat's little theorem: If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Review Questions

- Find $210 \text{ div } 17$ and $210 \text{ mod } 17$.
- Define what it means for a and b to be congruent modulo 7.
 - Which pairs of the integers $-11, -8, -7, -1, 0, 3$, and 17 are congruent modulo 7?
 - Show that if a and b are congruent modulo 7, then $10a + 13$ and $-4b + 20$ are also congruent modulo 7.
- Show that if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$.
- Describe a procedure for converting decimal (base 10) expansions of integers into hexadecimal expansions.
- Convert $(1101\ 1001\ 0101\ 1011)_2$ to octal and hexadecimal representations.
- Convert $(7206)_8$ and $(A0EB)_{16}$ to a binary representation.
- State the fundamental theorem of arithmetic.
- Describe a procedure for finding the prime factorization of an integer.
 - Use this procedure to find the prime factorization of $80,707$.
- Define the greatest common divisor of two integers.
 - Describe at least three different ways to find the greatest common divisor of two integers. When does each method work best?
 - Find the greatest common divisor of $1,234,567$ and $7,654,321$.
 - Find the greatest common divisor of $2^3 3^5 5^7 7^9 11$ and $2^9 3^7 5^5 7^3 13$.
- How can you find a linear combination (with integer coefficients) of two integers that equals their greatest common divisor?
 - Express $\gcd(84, 119)$ as a linear combination of 84 and 119 .
- What does it mean for $\bar{a}$ to be an inverse of a modulo m ?
 - How can you find an inverse of a modulo m when m is a positive integer and $\gcd(a, m) = 1$?
 - Find an inverse of 7 modulo 19 .
- How can an inverse of a modulo m be used to solve the congruence $ax \equiv b \pmod{m}$ when $\gcd(a, m) = 1$?
 - Solve the linear congruence $7x \equiv 13 \pmod{19}$.
- State the Chinese remainder theorem.
 - Find the solutions to the system $x \equiv 1 \pmod{4}$, $x \equiv 2 \pmod{5}$, and $x \equiv 3 \pmod{7}$.
- Suppose that $2^{n-1} \equiv 1 \pmod{n}$. Is n necessarily prime?
- Use Fermat's little theorem to evaluate $9^{200} \text{ mod } 19$.
- Explain how the check digit is found for a 10-digit ISBN.
- Encrypt the message APPLES AND ORANGES using a shift cipher with key $k = 13$.
- What is the difference between a public key and a private key cryptosystem?
 - Explain why using shift ciphers is a private key system.
 - Explain why the RSA cryptosystem is a public key system.
- Explain how encryption and decryption are done in the RSA cryptosystem.
- Describe how two parties can share a secret key using the Diffie-Hellman key exchange protocol.

Supplementary Exercises

- The odometer on a car goes to up 100,000 miles. The present owner of a car bought it when the odometer read 43,179 miles. He now wants to sell it; when you examine the car for possible purchase, you notice that the odometer reads 89,697 miles. What can you conclude about how many miles he drove the car, assuming that the odometer always worked correctly?
- Explain why $n \text{ div } 7$ equals the number of complete weeks in n days.
 - Explain why $n \text{ div } 24$ equals the number of complete days in n hours.
- Find four numbers congruent to 5 modulo 17.
- Show that if a and d are positive integers, then there are integers q and r such that $a = dq + r$, where $-d/2 < r \leq d/2$.
- * Show that if $ac \equiv bc \pmod{m}$, where a, b, c , and m are integers with $m > 2$, and $d = \gcd(m, c)$, then $a \equiv b \pmod{m/d}$.
- Show that the sum of the squares of two odd integers cannot be the square of an integer.
- Show that if $n^2 + 1$ is a perfect square, where n is an integer, then n is even.

8. Prove that there are no solutions in integers x and y to the equation $x^2 - 5y^2 = 2$. [Hint: Consider this equation modulo 5.]
9. Develop a test for divisibility of a positive integer n by 8 based on the binary expansion of n .
10. Develop a test for divisibility of a positive integer n by 3 based on the binary expansion of n .
11. Devise an algorithm for guessing a number between 1 and $2^n - 1$ by successively guessing each bit in its binary expansion.
12. Determine the complexity, in terms of the number of guesses, needed to determine a number between 1 and $2^n - 1$ by successively guessing the bits in its binary expansion.
13. Show that an integer is divisible by 9 if and only if the sum of its decimal digits is divisible by 9.
- **14. Show that if a and b are positive irrational numbers such that $1/a + 1/b = 1$, then every positive integer can be uniquely expressed as either $[ka]$ or $[kb]$ for some positive integer k .
15. Prove there are infinitely many primes by showing that $Q_n = n! + 1$ must have a prime factor greater than n whenever n is a positive integer.
16. Find a positive integer n for which $Q_n = n! + 1$ is not prime.
17. Use Dirichlet's theorem, which states there are infinitely many primes in every arithmetic progression $ak + b$ where $\gcd(a, b) = 1$, to show that there are infinitely many primes that have a decimal expansion ending with a 1.
18. Prove that if n is a positive integer such that the sum of the divisors of n is $n + 1$, then n is prime.
- *19. Show that every integer greater than 11 is the sum of two composite integers.
20. Find the five smallest consecutive composite integers.
21. Show that Goldbach's conjecture, which states that every even integer greater than 2 is the sum of two primes, is equivalent to the statement that every integer greater than 5 is the sum of three primes.
22. Find an arithmetic progression of length six beginning with 7 that contains only primes.
- *23. Prove that if $f(x)$ is a nonconstant polynomial with integer coefficients, then there is an integer y such that $f(y)$ is composite. [Hint: Assume that $f(x_0) = p$ is prime. Show that p divides $f(x_0 + kp)$ for all integers k . Obtain a contradiction of the fact that a polynomial of degree n , where $n > 1$, takes on each value at most n times.]
- *24. How many zeros are at the end of the binary expansion of $(100_{10})!$?
25. Use the Euclidean algorithm to find the greatest common divisor of 10,223 and 33,341.
26. How many divisions are required to find $\gcd(144, 233)$ using the Euclidean algorithm?
27. Find $\gcd(2n + 1, 3n + 2)$, where n is a positive integer. [Hint: Use the Euclidean algorithm.]
28. a) Show that if a and b are positive integers with $a \geq b$, then $\gcd(a, b) = a$ if $a = b$, $\gcd(a, b) = 2 \gcd(a/2, b/2)$ if a and b are even, $\gcd(a, b) = \gcd(a/2, b)$ if a is even and b is odd, and $\gcd(a, b) = \gcd(a - b, b)$ if both a and b are odd.
b) Explain how to use part (a) to construct an algorithm for computing the greatest common divisor of two positive integers that uses only comparisons, subtractions, and shifts of binary expansions, without using any divisions.
c) Find $\gcd(1202, 4848)$ using this algorithm.
29. Adapt the proof that there are infinitely many primes (Theorem 3 in Section 4.3) to show that there are infinitely many primes in the arithmetic progression $6k + 5$, $k = 1, 2, \dots$
30. Explain why you cannot directly adapt the proof that there are infinitely many primes (Theorem 3 in Section 4.3) to show that there are infinitely many primes in the arithmetic progression $3k + 1$, $k = 1, 2, \dots$
31. Explain why you cannot directly adapt the proof that there are infinitely many primes (Theorem 3 in Section 4.3) to show that there are infinitely many primes in the arithmetic progression $4k + 1$, $k = 1, 2, \dots$
32. Show that if the smallest prime factor p of the positive integer n is larger than $\sqrt[3]{n}$, then n/p is prime or equal to 1.
A set of integers is called **mutually relatively prime** if the greatest common divisor of these integers is 1.
33. Determine whether the integers in each of these sets are mutually relatively prime.
a) 8, 10, 12 b) 12, 15, 25
c) 15, 21, 28 d) 21, 24, 28, 32
34. Find a set of four mutually relatively prime integers such that no two of them are relatively prime.
- *35. For which positive integers n is $n^4 + 4^n$ prime?
36. Show that the system of congruences $x \equiv 2 \pmod{6}$ and $x \equiv 3 \pmod{9}$ has no solutions.
37. Find all solutions of the system of congruences $x \equiv 4 \pmod{6}$ and $x \equiv 13 \pmod{15}$.
- *38. a) Show that the system of congruences $x \equiv a_1 \pmod{m_1}$ and $x \equiv a_2 \pmod{m_2}$, where a_1, a_2, m_1 , and m_2 are integers with $m_1 > 0$ and $m_2 > 0$, has a solution if and only if $\gcd(m_1, m_2) \mid a_1 - a_2$.
b) Show that if the system in part (a) has a solution, then it is unique modulo $\text{lcm}(m_1, m_2)$.
39. Prove that 30 divides $n^9 - n$ for every nonnegative integer n .
40. Prove that $n^{12} - 1$ is divisible by 35 for every integer n for which $\gcd(n, 35) = 1$.
41. Show that if p and q are distinct prime numbers, then $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$.
The check digit a_{13} for an ISBN-13 with initial digits $a_1 a_2 \dots a_{12}$ is determined by the congruence $(a_1 + a_3 + \dots + a_{13}) + 3(a_2 + a_4 + \dots + a_{12}) \equiv 0 \pmod{10}$.

42. Determine whether each of these 13-digit numbers is a valid ISBN-13.
- 978-0-073-20679-1
 - 978-0-45424-521-1
 - 978-3-16-148410-0
 - 978-0-201-10179-9
43. Show that the check digit of an ISBN-13 can always detect a single error.
44. Show that there are transpositions of two digits that are not detected by an ISBN-13.
- A **routing transit number (RTN)** is a bank code used in the United States which appears on the bottom of checks. The most common form of an RTN has nine digits, where the last digit is a check digit. If $d_1 d_2 \dots d_9$ is a valid RTN, the congruence $3(d_1 + d_4 + d_7) + 7(d_2 + d_5 + d_8) + (d_3 + d_6 + d_9) \equiv 0 \pmod{10}$ must hold.
45. Show that if $d_1 d_2 \dots d_9$ is a valid RTN, then $d_9 = 7(d_1 + d_4 + d_7) + 3(d_2 + d_5 + d_8) + 9(d_3 + d_6) \pmod{10}$. Furthermore, use this formula to find the check digit that follows the eight digits 11100002 in a valid RTN.
46. Show that the check digit of an RTN can detect all single errors and determine which transposition errors an RTN check digit can catch and which ones it cannot catch.
47. The encrypted version of a message is LJMKG MG-MXF QEXMW. If it was encrypted using the affine cipher $f(p) = (7p + 10) \pmod{26}$, what was the original message?

Autokey ciphers are ciphers where the n th letter of the plaintext is shifted by the numerical equivalent of the n th letter of a keystream. The keystream begins with a seed letter; its subsequent letters are constructed using either the plaintext or the ciphertext. When the plaintext is used, each character of the keystream, after the first, is the previous letter of the plaintext. When the ciphertext is used, each subsequent character of the keystream, after the first, is the previous letter of the ciphertext computed so far. In both cases, plaintext letters are encrypted by shifting each character by the numerical equivalent of the corresponding keystream letter.

48. Use the autokey cipher to encrypt the message NOW IS THE TIME TO DECIDE (ignoring spaces) using
- the keystream with seed X followed by letters of the plaintext.
 - the keystream with seed X followed by letters of the ciphertext.
49. Use the autokey cipher to encrypt the message THE DREAM OF REASON (ignoring spaces) using
- the keystream with seed X followed by letters of the plaintext.
 - the keystream with seed X followed by letters of the ciphertext.

Computer Projects

Write programs with these inputs and outputs.

- Given integers n and b , each greater than 1, find the base b expansion of this integer.
- Given the positive integers a , b , and m with $m > 1$, find $a^b \pmod{m}$.
- Given a positive integer, find the Cantor expansion of this integer (see the preamble to Exercise 54 of Section 4.2).
- Given a positive integer, determine whether it is prime using trial division.
- Given a positive integer, find the prime factorization of this integer.
- Given two positive integers, find their greatest common divisor using the Euclidean algorithm.
- Given two positive integers, find their least common multiple.
- Given positive integers a and b , find Bézout coefficients s and t of a and b .
- Given relatively prime positive integers a and b , find an inverse of a modulo b .
- Given n linear congruences modulo pairwise relatively prime moduli, find the simultaneous solution of these congruences modulo the product of these moduli.
- Given a positive integer N , a modulus m , a multiplier a , an increment c , and a seed x_0 , where $0 \leq a < m$, $0 \leq c < m$, and $0 \leq x_0 < m$, generate the sequence of N pseudo-random numbers using the linear congruential generator $x_{n+1} = (ax_n + c) \pmod{m}$.
- Given a set of identification numbers, use a hash function to assign them to memory locations where there are k memory locations.
- Compute the check digit when given the first nine digits of an ISBN-10.
- Given a message and a positive integer k less than 26, encrypt this message using the shift cipher with key k ; and given a message encrypted using a shift cipher with key k , decrypt this message.
- Given a message and positive integers a and b less than 26 with $\gcd(a, 26)$, encrypt this message using an affine cipher with key (a, b) ; and given a message encrypted using the affine cipher with key (a, b) , decrypt this message, by first finding the decryption key and then applying the appropriate decryption transformation.
- Find the original plaintext message from the ciphertext message produced by encrypting the plaintext message